

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, JavaScript, C++, SQL

Tools and Frameworks: Tableau, Pandas, scikit-learn, OpenCV, PyTorch, LangChain, Pinecone, PostgreSQL, AWS, SAP, Redis, Kafka, Spark, Docker, Kubernetes, GitHub Actions(CI/CD), Git, Linux, FastAPI, React

Areas of Interest and Coursework: Data Processing pipelines, AI & LLMs, Software Engineering(SDLC)

INDUSTRY EXPERIENCE

Data AI Engineer | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Built a priority-ordered fallback in the **RAG chat agent** across 5 embedding models (OpenAI, Anthropic, Ollama), re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising evidence-backed control assessment **accuracy from 64% to 92%**.
- Diagnosed an out-of-memory failure in that fallback path, tracing it to freshly pulled 4-5 GB model weights lingering in the OS page cache, and fixed it with targeted **posix_fadvise** eviction, eliminating model-load failures.
- Developed backend services for an AI-native GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built an **AI-powered** PRD review platform in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.
- Architected a fully serverless review pipeline on AWS (S3 + CloudFront-hosted React SPA, Django on Lambda, DynamoDB, and an SQS-triggered Lambda AI worker) that retries failed analyses off a dead-letter queue and logs per-report model API spend, sustaining **200+** submissions across 5 cohorts at **~\$3/month** idle cost.

Software Development Engineer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Built the online ordering and booking platform by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to **Kafka** workers for order confirmation, customer notification, and crew allocation, taking **\$600K+** in online bookings across ~1,100 orders.
- Designed and deployed the **AWS architecture** that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for ~10x seasonal peaks and load-tested to **~50 quote requests/sec at <120 ms p99** with zero-downtime rolling deploys.

ACADEMIC EXPERIENCE / PROJECTS

Graduate Assistant | UMBC - Software Engineering & Data Structures

Jan 2025 – June 2026

- Developed 200+ unit tests for the DSA projects on Linked Lists, Self Balancing Trees, Heaps, and Hash Tables automating logic checking and reducing grading time by over **80%**.
- Helped 50+ students design and debug C++ projects implementing core data structures, and graded their homework.
- Mentored 10 student **Agile** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.

When Pixels Talk Back | Watermarking EMR Data to analyse impact on CV models

- Fine-tuned **3 computer-vision models** (ResNet50, DenseNet169, MobileNetV2) for classification over COVID-19 lung CT scans via two-stage ImageNet transfer learning, to establish a baseline peaking at **93%** accuracy and **0.951** F1.
- Benchmarked **3 watermarking algorithms** across all 3 models, showing the fragile scheme costs **0%** accuracy loss, versus ~1% for reversible embedding with a 300-bit payload and 11.2% for robust embedding.
- Embedded a 1024-bit blind fragile watermark into the Region-of-Non-Interest of CT scans via key-scrambled LSB embedding of a SHA-512 fingerprint, enabling content authentication with near-lossless quality at **SSIM 0.999**.